

TI (TI I). Strongest classified transitions below 2 μm (NIST): 12 levels, 18 lines.

The diagram illustrates the energy levels and transitions for the $2p^0$, $2D$, and $2S$ systems. The energy levels are represented by horizontal lines, and the transitions are indicated by colored arrows with associated wavelength and energy values.

Energy Levels:

- $8p \ ^2P_{3/2}^o$: 41740.800 cm^{-1} , $\tau = 128 \text{ ns}$
- $8p \ ^2P_{1/2}^o$: 41368.100 cm^{-1} , $\tau = 184 \text{ ns}$
- $7p \ ^2P_{3/2}^o$: 35161.100 cm^{-1} , $\tau = 48 \text{ ns}$
- $7p \ ^2P_{1/2}^o$: 34159.900 cm^{-1} , $\tau = 63.1 \text{ ns}$
- $6p \ ^2P_{3/2}^o$: 7792.700 cm^{-1} , $\tau = 2.66 \text{ ns}$
- $6p \ ^2P_{1/2}^o$: 0.000 cm^{-1} , stable
- $7s \ ^2S_{1/2}$: 26477.500 cm^{-1} , $\tau = 7.45 \text{ ns}$
- $8s \ ^2S_{1/2}$: 38745.900 cm^{-1}
- $6d \ ^2D_{5/2}$: 36199.900 cm^{-1}
- $6d \ ^2D_{3/2}$: 36117.900 cm^{-1} , $\tau = 6.2 \text{ ns}$
- $7d \ ^2D_{5/2}$: 42049.000 cm^{-1}
- $7d \ ^2D_{3/2}$: 42011.400 cm^{-1}

Transitions:

- Red Arrows:**
 - $8p \ ^2P_{3/2}^o \rightarrow 8p \ ^2P_{1/2}^o$: 655.166 nm | 1.855 eV
 - $8p \ ^2P_{1/2}^o \rightarrow 7p \ ^2P_{3/2}^o$: 671.565 nm | 1.052 eV
 - $8p \ ^2P_{1/2}^o \rightarrow 7p \ ^2P_{1/2}^o$: 671.565 nm | 1.052 eV
 - $8p \ ^2P_{1/2}^o \rightarrow 6p \ ^2P_{3/2}^o$: 671.565 nm | 1.052 eV
 - $8p \ ^2P_{1/2}^o \rightarrow 6p \ ^2P_{1/2}^o$: 671.565 nm | 1.052 eV
 - $7p \ ^2P_{3/2}^o \rightarrow 7s \ ^2S_{1/2}$: 313.068 nm | 0.759 eV
 - $7p \ ^2P_{3/2}^o \rightarrow 8s \ ^2S_{1/2}$: 258.092 nm | 0.553 eV
 - $7p \ ^2P_{1/2}^o \rightarrow 7s \ ^2S_{1/2}$: 313.068 nm | 0.759 eV
 - $7p \ ^2P_{1/2}^o \rightarrow 8s \ ^2S_{1/2}$: 258.092 nm | 0.553 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 6d \ ^2D_{5/2}$: 313.068 nm | 0.759 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 6d \ ^2D_{3/2}$: 258.092 nm | 0.553 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 7s \ ^2S_{1/2}$: 313.068 nm | 0.759 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 8s \ ^2S_{1/2}$: 258.092 nm | 0.553 eV
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 - $6p \ ^2P_{3/2}^o \rightarrow 8d \ ^2D_{5/2}$: 313.068 nm | 0.759 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 8d \ ^2D_{3/2}$: 258.092 nm | 0.553 eV
- Green Arrows:**
 - $6p \ ^2P_{3/2}^o \rightarrow 7s \ ^2S_{1/2}$: 313.068 nm | 0.759 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 8s \ ^2S_{1/2}$: 258.092 nm | 0.553 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 7d \ ^2D_{5/2}$: 313.068 nm | 0.759 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 7d \ ^2D_{3/2}$: 258.092 nm | 0.553 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 8d \ ^2D_{5/2}$: 313.068 nm | 0.759 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 8d \ ^2D_{3/2}$: 258.092 nm | 0.553 eV
- Blue Arrows:**
 - $6p \ ^2P_{3/2}^o \rightarrow 6d \ ^2D_{5/2}$: 313.068 nm | 0.759 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 6d \ ^2D_{3/2}$: 258.092 nm | 0.553 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 7s \ ^2S_{1/2}$: 313.068 nm | 0.759 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 8s \ ^2S_{1/2}$: 258.092 nm | 0.553 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 7d \ ^2D_{5/2}$: 313.068 nm | 0.759 eV
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 - $6p \ ^2P_{3/2}^o \rightarrow 8d \ ^2D_{5/2}$: 313.068 nm | 0.759 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 8d \ ^2D_{3/2}$: 258.092 nm | 0.553 eV
- Black Arrows:**
 - $6p \ ^2P_{3/2}^o \rightarrow 6d \ ^2D_{5/2}$: 313.068 nm | 0.759 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 6d \ ^2D_{3/2}$: 258.092 nm | 0.553 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 7s \ ^2S_{1/2}$: 313.068 nm | 0.759 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 8s \ ^2S_{1/2}$: 258.092 nm | 0.553 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 7d \ ^2D_{5/2}$: 313.068 nm | 0.759 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 7d \ ^2D_{3/2}$: 258.092 nm | 0.553 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 8d \ ^2D_{5/2}$: 313.068 nm | 0.759 eV
 - $6p \ ^2P_{3/2}^o \rightarrow 8d \ ^2D_{3/2}$: 258.092 nm | 0.553 eV

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e J' \rangle $ (ea)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$6p\ ^2P_{3/2}^o - 6d\ ^2D_{5/2}$	352.0234	351.9228	851.6264	4.002	1.240e+08	-	-	352 nm proposed laser-cooling transition 6P3/2
$6p\ ^2P_{3/2}^o - 7s\ ^2S_{1/2}$	377.6780	377.5707	793.777941	1.823	6.250e+07	21.36	46.3	377.6 nm resonance line
$6p\ ^2P_{3/2}^o - 7s\ ^2S_{1/2}$	535.1944	535.0456	560.1562	3.266	7.050e+07	21.36	53.7	535.0 nm green line
$7s\ ^2S_{1/2} - 8p\ ^2P_{3/2}^o$	655.1663	654.9854	457.5822	1.855	6.200e+06	1.246	79.2	655 nm second step of two-step spectroscopy
$7s\ ^2S_{1/2} - 8p\ ^2P_{1/2}^o$	671.5646	671.3793	446.4090	1.052	3.700e+06	0.8645	68.1	671 nm second step of two-step spectroscopy
$6p\ ^2P_{3/2}^o - 6p\ ^2P_{1/2}^o$	1283.2523	1282.9013	233.6193	-	4.094e+00	6.735e-07	96.7	1283 nm M1
$7s\ ^2S_{1/2} - 7p\ ^2P_{1/2}^o$	1301.6766	1301.3206	230.3126	6.101	1.710e+07	2.522	100	1301 nm second step of two-step

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$6p\ ^2P_{3/2}^o - 7s\ ^2S_{1/2}$	377.678	205-203	1658.3(7)
$7s\ ^2S_{1/2} - 8p\ ^2P_{3/2}^o$	655.166	203-205	463.4(7)
$7s\ ^2S_{1/2} - 8p\ ^2P_{1/2}^o$	671.565	203-205	450.1(8)
$7s\ ^2S_{1/2} - 7p\ ^2P_{1/2}^o$	1301.677	203-205	534.4(9)

Arrow label: vacuum wavelength reduced dipole matrix element $\langle V \| e r \| V \rangle$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model or semi-empirical calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$6p\ ^2P_{3/2}^o$	21310.835(5)	-	F=0: -15983.13 F=1: +5327.71
$6p\ ^2P_{3/2}^o$	265.03830(10)	-	F=1: -331.30 F=2: +198.78
$7s\ ^2S_{1/2}$	12296.1(7)	-	F=0: -9222.07 F=1: +3074.02
$7p\ ^2P_{1/2}^o$	2173.3(8)	-	F=0: -1629.97 F=1: +543.33
$7p\ ^2P_{3/2}^o$	311.4(3)	-	F=1: -389.25 F=2: +233.55
$6d\ ^2D_{3/2}$	-42.9(4)	-	F=1: +53.62 F=2: -32.17
$6d\ ^2D_{5/2}$	226.9(3)	-	F=2: -397.07 F=3: +283.62
$8s\ ^2S_{1/2}$	3870.8(14)	-	F=0: -2903.10 F=1: +967.70
$8p\ ^2P_{3/2}^o$	788.5(6)	-	F=0: -591.38 F=1: +197.12
$8p\ ^2P_{3/2}^o$	131.69(20)	-	F=1: -164.61 F=2: +98.77
$7d\ ^2D_{3/2}$	-56.3(6)	-	F=1: +70.38 F=2: -42.23
$7d\ ^2D_{5/2}$	180.20(20)	-	F=2: -315.35 F=3: +225.25

Measured A, B constants; sources in the data file.